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Diverse Minds Think Alike: A SHAP-Guided RNN for Stock Price Forecasting with Smart SMA Optimization

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***Abstract*—Stock market forecasting is challenging due to volatility, complex temporal dependencies and nonlinear patterns. Many models underutilize features, lack interpretability and fail to leverage model diversity in ensembles. This paper proposes an explainable ensemble framework combining attention enhanced recurrent networks (LSTM, GRU, BiLSTM) with engineered technical and sentiment features. Optimal Simple Moving Average (SMA) windows are selected via a hybrid A* search and dynamic programming method. SHapley Additive exPlanations (SHAP) quantify feature importance and inter model diversity, guiding a dynamic weighting scheme for ensemble aggregation. These weighted outputs along with base model predictions are input to an XGBoost meta-learner for final forecasting. Experiments on real world stock and sentiment data show the method performs better than individual models and static ensembles achieving an MSE of 1.0445, MAE of 0.6397 and R^2 of 0.9861. The framework advances explainability, temporal modeling and diversity aware ensembling for more reliable stock price prediction.**

***Index Terms*—Stock forecasting, Reinforcement learning, Dynamic programming, A* search, Technical indicators, SMA optimization**

I. Introduction

Stock price forecasting remains difficult due to volatility, non-linearity and external market influences. Traditional statistical and machine learning models struggle to capture the temporal dependencies and contextual signals essential for accurate prediction. Deep learning methods particularly RNN variants such as LSTM and GRU has better model sequential patterns but still face key limitations.

Most approaches focus on price based technical indicators while neglecting market sentiment and index dynamics. Ensemble models often use static weights and overlook diversity among base learners while interpretability is rarely addressed which limits trust in high stakes financial contexts.

We propose an attention enhanced ensemble of LSTM, GRU, and BiLSTM models. SMA parameters for technical indicators are optimized using a hybrid A* dynamic programming method. SHAP values quantify feature importance and model diversity, guiding adaptive weighting with final predictions refined by an XGBoost meta learner. The framework delivers both high accuracy and interpretability making it suitable for applications such as algorithmic trading, portfolio optimization and regulatory risk assessment.

II. Related Work

A. Deep Sequential Models in Stock Prediction

Recurrent neural networks (RNNs), particularly Long Short-Term Memory (LSTM) and Gated Recurrent Units (GRU), are widely used to capture temporal dependencies in financial time series. Gülmez [1] enhanced LSTM accuracy with the Artificial Rabbits Optimization algorithm, while Saeed and Yin [22] compared LSTM, GRU, and CNN for South Asian markets, emphasizing architecture choice. Ji et al. [10] explored deep learning architectures for capturing complex temporal dependencies in stock data. Bidirectional LSTM (BiLSTM) extends context learning to both past and future inputs. Wang [9] coupled BiLSTM with an improved Transformer, and Liu [11] incorporated attention to refine temporal focus. Gülmez [18] further combined attention with fuzzy logic for enhanced prediction. However, most works evaluate these models in isolation without mechanisms to combine their complementary strengths.

B. Attention Mechanisms and Transformer Based Architectures

Attention mechanisms improve temporal feature selection. Liu [4] used attentive temporal convolutions in GANs, and Li et al. [19] proposed a GAN-LSTM-Attention framework. Lu [17] and Zhou et al. [16] integrated attention with BiLSTM and multimodal architectures. Li and Xu [20] enhanced prediction using GANs with Transformer based attention. Even with strong results, most approaches prioritize accuracy over transparency, rarely interpreting feature importance or integrating attention into ensembles or meta learning pipelines.

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C. Graph-Based and Multimodal Learning Approaches

Graph and multimodal learning capture market structure and context. Amiri et al. [8] used a GCN-LSTM for temporal–relational patterns, and Mehrabian [6] combined graph learning with state-space modeling. Karadaş et al. [7] fused textual, numerical, and visual data, while Wu et al. [21] combined CNN-LSTM with economic indicators. These methods broaden modeling diversity but often require heavy feature engineering and offer limited interpretability. Their ensemble potential remains underexplored.

D. Ensemble and Hybrid Learning Strategies

Ensemble learning combines diverse predictors for improved performance. Chaudhari and Mahajan [3] incorporated sentiment analysis, and Beniwal et al. [12] applied ensemble regressors for long-horizon forecasts. Li and Shi [14] optimized neural inputs, while Zhu et al. [15] used Holt-Winters smoothing and PCA whitening. Wang et al. [13] proposed a scalable AI-cloud-based framework for stock forecasting. However, most ensembles rely on static weights, overlook redundancy among models and seldom employ diversity metrics, interpretability tools or advanced meta learners such as stacking.

E. Emerging Architectures: LLMs and State Space Models

Recent works have introduced large language models (LLMs) and state space architectures into financial forecasting. Wang [5] developed StockTime for time series prediction, Zhou et al. [16] explored LLM-augmented Transformer–CNN hybrids, and Chang et al. [2] with Mehrabian [6] proposed the Mamba architecture for efficient long range dependency modeling.

Across prior works, several key limitations emerge. Most deep learning and ensemble models fail to explicitly quantify and utilize prediction diversity, leading to correlated and redundant ensemble decisions. Interpretability tools like SHAP are rarely integrated into ensemble pipelines to explain and refine model behavior. While stacking offers a powerful meta learning strategy, its application in financial time series forecasting remains limited. We address these by proposing a SHAP driven ensemble of attention enhanced LSTM, GRU, and BiLSTM models. SHAP value correlations guide dynamic weighting to promote complementary behaviors, while base and ensemble outputs form meta features for an XGBoost meta learner. This unified approach delivers diversity aware, interpretable and high performance stock prediction.

III. Methodology

This section presents the detailed methodology of our proposed explainable ensemble forecasting framework. The overall workflow is illustrated in Fig. 1.

A. Data Collection

This study utilizes data from the Dhaka Stock Exchange (DSE) from January 2023 to May 2025, comprising:

- Stock Price Data: Historical records (Open, High, Low, Close, Volume) of selected companies.
- Market Summary Data: Daily DSEX/DS30 indices, market capitalization, and turnover.
- News Data: Financial news articles aligned with trading dates to extract sentiment.

B. Data Preprocessing

A systematic pipeline was implemented to transform raw data into inputs suitable for model training.

1) Sequential Sorting and Windowing: The dataset was chronologically sorted and segmented using a 30-day sliding window, producing sequential samples.

2) Feature Engineering: To enrich the feature set, multiple financial indicators and lagged features were computed.

a) Basic Indicators: The following key indicators were calculated for each day t:

$$\text{Return}_t = \frac{\text{CLOSEP}_t - \text{YCP}_t}{\text{YCP}_t}, \tag{1}$$

$$\text{Gap}_t = \frac{\text{OPENP}_t - \text{YCP}_t}{\text{YCP}_t}, \tag{2}$$

$$\text{Intraday Change}_t = \frac{\text{CLOSEP}_t - \text{OPENP}_t}{\text{OPENP}_t}. \tag{3}$$

b) Technical Indicators and Lag Features: Additional features capturing price trends, volatility, and momentum were extracted.

c) SMA Optimization Using A and Dynamic Programming:* We optimize SMA window sizes $\mathbf{w} = \{w_1, \ldots, w_k\}$ by maximizing a validation metric $\mathcal{M}(\mathbf{w})$ over candidate windows. Due to the combinatorial search space, we employ A* search with a heuristic estimate $h(\mathbf{w})$, while dynamic programming caches evaluated subsets:

$$\mathbf{w} = \arg\max_{\mathbf{w} \subseteq \mathcal{W}} \mathcal{M}(\mathbf{w}) \tag{4}$$

- Simple Moving Averages (SMA):

$$\text{SMA}_{19,t} = \frac{1}{19} \sum_{i=0}^{18} \text{CLOSEP}_{t-i}, \tag{5}$$

$$\text{SMA}_{47,t} = \frac{1}{47} \sum_{i=0}^{46} \text{CLOSEP}_{t-i} \tag{6}$$

- Volatility:

$$\text{Volatility}_t = \text{std}(\text{Return}_{t-4}, \ldots, \text{Return}_t) \tag{7}$$

- Price Gap:

$$\text{Price Gap}_t = \text{HIGH}_t - \text{LOW}_t \tag{8}$$

- Momentum:

$$\text{Momentum}_{5,t} = \text{CLOSEP}_t - \text{CLOSEP}_{t-5} \tag{9}$$

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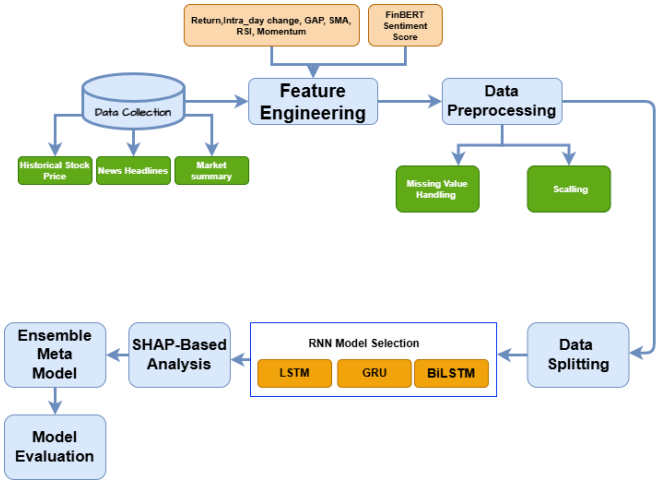


Fig. 1. Overall workflow of the proposed method.

- Relative Strength Index (RSI):

$$\Delta_t = \text{CLOSEP}_t - \text{CLOSEP}_{t-1} \tag{10}$$

$$\text{Gain}_t = \max(\Delta_t, 0), \quad \text{Loss}_t = \max(-\Delta_t, 0) \tag{11}$$

$$\text{AvgGain}_t = \text{mean}(\text{Gain}_{t-13}, \ldots, \text{Gain}_t) \tag{12}$$

$$\text{AvgLoss}_t = \text{mean}(\text{Loss}_{t-13}, \ldots, \text{Loss}_t) \tag{13}$$

$$\text{RS}_t = \frac{\text{AvgGain}_t}{\text{AvgLoss}_t + \epsilon}, \quad \epsilon = 10^{-9} \tag{14}$$

$$\text{RSI}_t = 100 - \frac{100}{1 + \text{RS}_t} \tag{15}$$

C. Sentiment Scoring with FinBERT

Market sentiment was quantified using FinBERT, a BERT based model fine tuned on financial texts. For each day t, sentiment scores were computed:

$$[P_{\text{pos}}, P_{\text{neu}}, P_{\text{neg}}] = \text{softmax}(W \cdot h_{[\text{CLS}]} + b) \tag{16}$$

$$S_t = P_{\text{pos}} - P_{\text{neg}}, \quad S_t \in [-1, 1] \tag{17}$$

Daily scores were averaged; missing days were assigned $S_t = 0$.

1) Train-Test Split and Scaling: The dataset was split into training and testing sets in an 80:20 ratio. Min-Max scaling was applied only on the training set:

$$X_{\text{scaled}} = \frac{X - X_{\min}}{X_{\max} - X_{\min}} \tag{18}$$

$$y_{\text{scaled}} = \frac{y - y_{\min}}{y_{\max} - y_{\min}} \tag{19}$$

D. Model Architecture

The architecture integrates deep temporal models, SHAP guided weighting, and XGBoost-based meta-learning shown in Fig. 2.

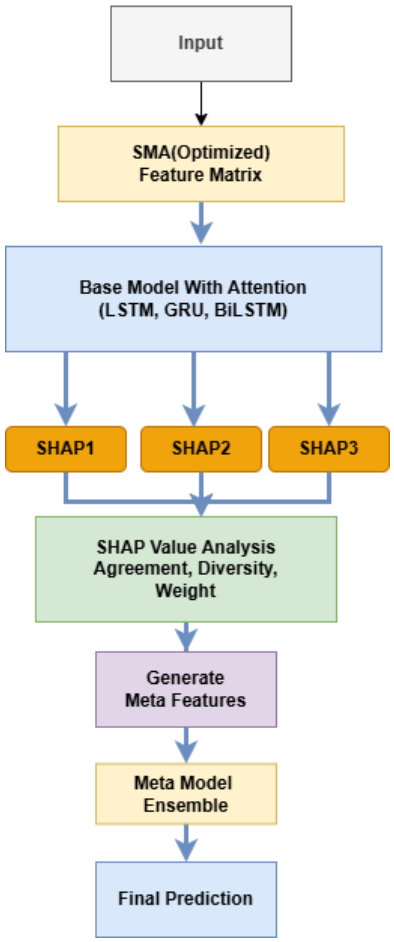


Fig. 2. Proposed ensemble model with SHAP-based weights and XGBoost meta-learner.

1) Rationale for Base Model Selection: We use LSTM, GRU, and BiLSTM due to their complementary strengths in sequence modeling:

- LSTM: Captures long-term dependencies.
- GRU: Faster and fewer parameters, suitable for short-term patterns.
- BiLSTM: Uses forward and backward context, improving generalization.

2) SHAP Based Diversity Weighting: For each model M_i, SHAP value tensors $\mathbf{S}^{(i)} \in \mathbb{R}^{N \times T \times F}$ are flattened to vectors $\mathbf{s}^{(i)}$:

$$\mathbf{A}_{i,j} = \text{corr}(\mathbf{s}^{(i)}, \mathbf{s}^{(j)}) \tag{20}$$

$$\mathbf{D}_{i,j} = 1 - |\mathbf{A}_{i,j}| \tag{21}$$

$$d_i = \frac{1}{k} \sum_{j=1}^{k} \mathbf{D}_{i,j} \tag{22}$$

$$\tilde{w}_i = \frac{1}{d_i + \epsilon}, \quad w_i = \frac{\tilde{w}_i}{\sum_{j=1}^{k} \tilde{w}_j} \tag{23}$$

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3) Meta-Feature Construction: To facilitate ensemble learning, we first collect the predictions from each of the k base models, where $\hat{y}^{(i)} \in \mathbb{R}^N$ represents the prediction vector from the i-th model for N samples. These predictions are aggregated into a meta-feature matrix $\mathbf{Z} \in \mathbb{R}^{N \times k}$ as follows:

$$\mathbf{Z} = \begin{bmatrix} \hat{y}^{(1)} & \cdots & \hat{y}^{(k)} \end{bmatrix} \in \mathbb{R}^{N \times k} \tag{24}$$

A weighted ensemble prediction $\hat{y}^{(\text{ens})}$ is then computed using the contribution weights w_i assigned to each base model:

$$\hat{y}^{(\text{ens})} = \sum_{i=1}^{k} w_i \cdot \hat{y}^{(i)} \tag{25}$$

To enrich the representational capacity of the meta-learner, we concatenate the ensemble prediction $\hat{y}^{(\text{ens})}$ with the original meta-feature matrix $\mathbf{Z}$, forming the final meta-feature matrix $\mathbf{X}^{(\text{meta})}$:

$$\mathbf{X}^{(\text{meta})} = [\mathbf{Z} \mid \hat{y}^{(\text{ens})}] \tag{26}$$

This augmented feature representation enables the meta-learner to exploit both individual model predictions and their combined consensus, improving overall predictive performance.

4) Meta-Ensemble Learning: Final prediction is made using an XGBoost regressor f_{meta}:

$$\hat{y}^{(\text{final})} = f_{\text{meta}}(\mathbf{X}^{(\text{meta})}) \tag{27}$$

IV. Results and Analysis

This section presents an in depth evaluation of the proposed SHAP-guided ensemble forecasting framework.

A. Evaluation of Base Models

Trained on shared technical, momentum, and sentiment features, BiLSTM achieved the best MSE, MAE, and R^2 as shown in Table I, followed by GRU with faster convergence and LSTM with stable performance. Their complementary strengths support ensemble integration.

TABLE I
Performance of Base Models

Model	MSE	MAE	R^2
LSTM	1.416	0.872	0.9632
GRU	1.350	0.849	0.9678
BiLSTM	**1.312**	**0.822**	**0.9699**

B. SHAP-Based Feature Attribution and Model Interpretation

To enhance interpretability, SHAP (SHapley Additive exPlanations) values were computed for each model, quantifying each feature's marginal contribution to predictions.

For BiLSTM, DSEX Index, Sentiment, and DS30 Index were the top contributors, indicating strong influence of market wide indicators and sentiment on stock movements. GRU prioritized Last Trading Price (LTP), INTRADAY_CHANGE, and Yield Closing Price (YCP), reflecting reliance on short-term volatility. LSTM emphasized DS30 Index, HIGH, GAP, and Sentiment, showing a preference for static daily measures and broader trends.

These results confirm that, despite sharing the same inputs, the models capture distinct patterns—supporting the diversity-driven ensemble approach adopted later.

C. Diversity Quantification via SHAP Correlation Analysis

To capture complementary model behavior, SHAP value tensors were flattened, and pairwise Pearson correlations computed across models shown in Fig. 3. Correlations were converted to diversity scores, where lower values indicate greater interpretive uniqueness.

BiLSTM and GRU showed the lowest correlation, while LSTM exhibited moderate overlap with both, serving as a stabilizing intermediary. Adaptive weights, derived from diversity scores shown in Table II, assigned BiLSTM the highest weight for its strong performance and distinct perspective, enhancing ensemble effectiveness.This dynamic weighting is different from fixed ensembles because it values both accuracy and variety in how models think, leading to decisions that are both correct and well rounded.

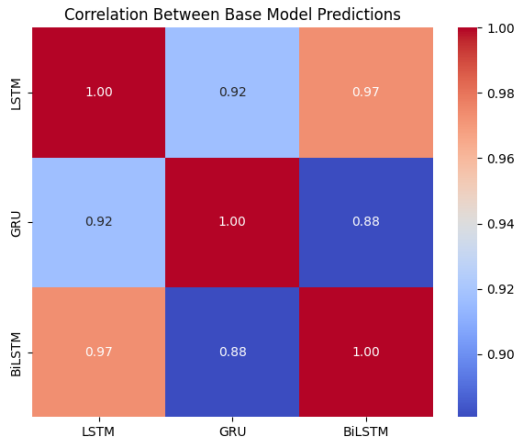


Fig. 3. SHAP correlation matrix between base models. Lower values indicate higher model diversity.

TABLE II
Model Weights from SHAP Diversity

Model	SHAP-Based Weight
LSTM	0.2847
GRU	0.2461
BiLSTM	**0.4692**

D. Performance of SHAP Guided Meta Ensemble

The final ensemble layer employs an XGBoost regressor using SHAP-weighted base model predictions and the ensemble average as inputs. As shown in Table III, the meta-learner surpasses all base models, achieving an R^2 of 0.9861 (explaining nearly 99% of target variance) and significantly reducing MSE and MAE.

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TABLE III
META-ENSEMBLE FINAL EVALUATION

Model	MSE	MAE	R^2
Meta-Ensemble (XGBoost)	**1.0445**	**0.6397**	**0.9861**

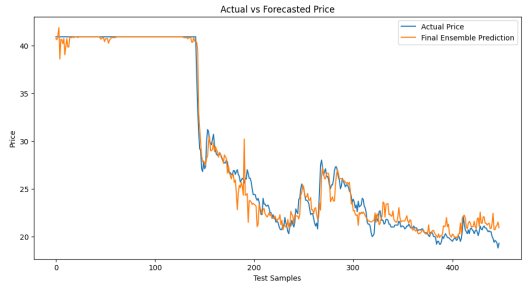


Fig. 4. Actual vs. Predicted Closing Prices on the Test Set.

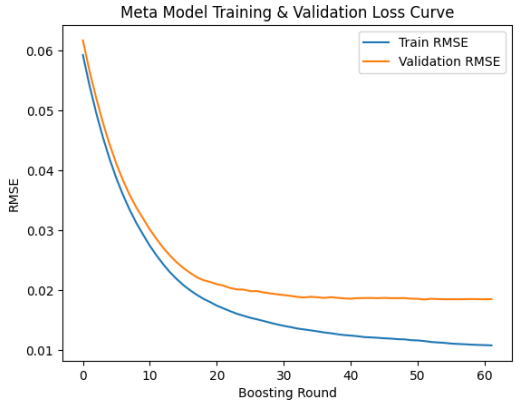


Fig. 5. RMSE Loss Curve during Training and Validation.

Fig. 4 shows predicted prices closely following actual values with minimal lag and high directional accuracy. RMSE loss curves shown in Fig. 5 demonstrate steady convergence without overfitting, confirming stability and strength under cross temporal testing.

The framework unites temporal modeling, interpretability and diversity aware ensembling. Recurrent models capture sequences, SHAP explains predictions and adaptive weighting promotes complementary behavior. Unlike static averaging, it uses diversity metrics for informed weighting, yielding an accurate, transparent system suited for trading and portfolio management. Its modular design enables use across exchanges with future work adding real time economic signals for greater strength.

V. CONCLUSION

Stock price prediction is hindered by market volatility and the limited interpretability of deep learning models. Many approaches neglect model diversity and explainability. This work presents a SHAP guided ensemble of attention based LSTM, GRU, and BiLSTM models, combined via diversity aware weighting and meta learning. On DSE data, the framework achieved an R^2 of 0.9861 while offering clear insights into feature importance and model roles. Future work will incorporate economic event data into the weighting scheme and extend the framework to other emerging markets to improve performance and generalizability.

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